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MODULE - 3

COUNTING TECHNIQUES

There are two main principles that are:

1. Sum Rule

Suppose an event can occur in m ways & a second event can occur in n ways and they can occur simultaneously then E or F occurs m+n ways.

Eg:

There are 5 female & 8 male profs handling Calculus subject. A student can choose how many ways?

no. of ways = 5+8 = 13

2. Product Rule

Suppose an event E can occur in m ways & F is another event can occur in n ways and it is independant. Then E and F can occur in m·n ways.

Eg:

Chairs of an auditorium are to be labelled with a letter and +ve integer less than 100. How many unique chairs?

no. of chairs = 26 X 100 = 2600

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* PIGEON HOLE PRINCIPLE

If there are 'p' pigeons and 'h' holes then $h < p$, such that atleast one hole contains atleast $\left\lceil \frac{p}{h} \right\rceil$ pigeons, i.e minimum number of pigeons.
└ finding lower bound

Note:

$$\left\lceil \frac{p}{h} \right\rceil = k$$

$$P = H(k-1) + 1$$

Ex:

1. Among 100 peoples atleast how many of them celebrating their bday in the same month of particular year?

$P = 100$; $H = 12$

each month $= \left\lceil \frac{100}{12} \right\rceil = 9$ //.

2.

$P = 367$ people
$H = 365$ days of a year

same day $= \left\lceil \frac{367}{365} \right\rceil = 2$ //.

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3. min num of students required in the class, to be secured that atleast 6 students will receive the same grade? If there are 5 grades.

$P = H(k-1) + 1$

here, $H = 5$; $K = 6$

$P = 5(6-1) + 1 = 25+1$
$= 26$ //

* PERMUTATION

Any arrangement of a set of n objects in a given order r, is called permutation.

$P(n, r) = \frac{n!}{(n-r)!} = {}^nP_r$ (without repetition)

$P(n, r, r_1, r_2 \ldots r_n) = \frac{n!}{r_1! \, r_2! \, r_3! \ldots}$ (with repetition)

eg: 6 objects, say : a, b, c, d, e, f, taken 3 at a time.

no. of ways $\{{}^6P_3\}$ $= \frac{6!}{(6-3)!} = \frac{6!}{3!} = 6 \times 5 \times 4 = 120$ //

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Ex.

1.

How many 7 letter-words can be formed using the letters of BENZENE.

3 E ; 2 N

$P(7, 3, 2) = \frac{7!}{3!\,2!} = \frac{7 \times 6 \times 5 \times 4 \times 2}{2} = 420$ //

2.

How many 5 letter words can be formed using letters of BABBY?

$P(5, 3) = \frac{5!}{3!} = 5 \times 4 = 20$ //

* COMBINATION

A combination of n objects taken r at a time is any selection r of the objects; where order does not count, then

$$C(n, r) = {}^{n}C_{r} = \frac{n!}{(n-r)!\; r!}$$

Eg:

4 objects : A, B, C, D taken 3 at a time.

$${}^{4}C_{3} = \frac{4!}{3!} = 4 \text{ //}$$

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Ex.

1\. How many committee of 3 can be formed from 8 people ?

no. of committees $= {}^8C_r = \frac{8!}{5!\,3!} = \frac{8\times7\times6}{6} = 56$ //

2\. A farmer buys 3 cows, 2 pigs & 4 hens from a man who has 6 cows, 5 pigs & 8 hens. How many choices does the farmer have ?

no. of choices $= {}^6C_3 \times {}^5C_2 \times {}^8C_4$

$= \frac{\cancel{6}\times5\times4}{\cancel{6}} \times 10 \times \frac{\cancel{8}^{2}\times7\times\cancel{6}\times5}{\cancel{4}\times\cancel{6}}$

$= 20 \times 20 \times 35$

$= 14000$ //

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* PRINCIPLE OF INCLUSION AND EXCLUSION

- Inclusion

$n(A\cup B) = [n(A) + n(B)] - n(A\cap B)$.

$n(A\cup B\cup C) = [n(A) + n(B) + n(C)] - [n(A\cap B) + n(B\cap C) + n(A\cap C)] + n(A\cap B\cap C)$.

X. $n(A\cup B\cup C\cup D) = [n(A) + n(B) + n(C) + n(D)] - [n(A\cap B) + n(A\cap C) + n(A\cap D) + n(B\cap C) + n(B\cap D) + n(C\cap D)] + [n(A\cap B\cap C) + n(A\cap B\cap D) + n(B\cap C\cap D) + n(A\cap D\cap C)] - n(A\cap B\cap C\cap D)$ //

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Ex.

1. Find no. of integers b/w 1-250 (both inclusive) that are divisible by any of the integers : 2, 3, 5, 7.

Let A be set of integers divisible by 2
B " by 3
C " by 5
D " by 7

$n(A) = \left\lfloor \frac{250}{2} \right\rfloor = 125$ $\quad$ $n(A \cap B) = \left\lfloor \frac{250}{6} \right\rfloor = 41$

$n(B) = \left\lfloor \frac{250}{3} \right\rfloor = 83$ $\quad$ $n(B \cap C) = \left\lfloor \frac{250}{15} \right\rfloor = 16$

$n(C) = \left\lfloor \frac{250}{5} \right\rfloor = 50$ $\quad$ $n(C \cap D) = \left\lfloor \frac{250}{35} \right\rfloor = 7$

$n(D) = \left\lfloor \frac{250}{7} \right\rfloor = 35$ $\quad$ $n(A \cap C) = \left\lfloor \frac{250}{10} \right\rfloor = 25$

$n(A \cap D) = \left\lfloor \frac{250}{14} \right\rfloor = 17$ $\quad$ $n(A \cap B \cap C) = \left\lfloor \frac{250}{6 \cdot 5} \right\rfloor = 8$

$n(B \cap D) = \left\lfloor \frac{250}{21} \right\rfloor = 11$ $\quad$ $n(A \cap B \cap D) = \left\lfloor \frac{250}{6 \times 7} \right\rfloor = 5$

$n(B \cap C \cap D) = \left\lfloor \frac{250}{15 \times 7} \right\rfloor = 2$ $\quad$ $n(A \cap C \cap D) = \left\lfloor \frac{250}{70} \right\rfloor = 3$

$n(A \cap B \cap C \cap D) = \left\lfloor \frac{250}{210} \right\rfloor = 1$

no. of integers $= (125 + 83 + 50 + 35) - (41 + 16 + 7 + 25 + 17 + 11) + (8 + 5 + 2 + 3) - 1$

$= 193$ //

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NOTE :

QN can be asked to find not divisible numbers too.
To find : n(universal) − n(divisible).

2.

Find the no. of the int <=1000 and not divisible by 3, 5, 7, 22.

Let A be set of int div by 3
Let B " 5
Let C " 7
Let D " 22

$n(A) = \left\lfloor \frac{1000}{3} \right\rfloor = 333$

$n(A \cap B) = \left\lfloor \frac{1000}{15} \right\rfloor = 66$

$n(B) = \left\lfloor \frac{1000}{5} \right\rfloor = 200$

$n(B \cap C) = \left\lfloor \frac{1000}{35} \right\rfloor = 28$

$n(C) = \left\lfloor \frac{1000}{7} \right\rfloor = 142$

$n(C \cap D) = \left\lfloor \frac{1000}{154} \right\rfloor = 6$

$n(D) = \left\lfloor \frac{1000}{22} \right\rfloor = 45$

$n(A \cap C) = \left\lfloor \frac{1000}{21} \right\rfloor = 47$

$n(A \cap D) = \left\lfloor \frac{1000}{66} \right\rfloor = 15$

$n(A \cap B \cap C) = \left\lfloor \frac{1000}{105} \right\rfloor = 9$

$n(B \cap D) = \left\lfloor \frac{1000}{110} \right\rfloor = 9$

$n(A \cap B \cap D) = \left\lfloor \frac{1000}{330} \right\rfloor = 3$

$n(A \cap C \cap D) = \left\lfloor \frac{1000}{462} \right\rfloor = 2$

$n(B \cap C \cap D) = \left\lfloor \frac{1000}{770} \right\rfloor = 1$

$n(A \cap B \cap C \cap D) = \left\lfloor \frac{1000}{1540} \right\rfloor = 0$

$\therefore n(A \cup B \cup C \cup D) = 564$ //

$\therefore n(\overline{A \cup B \cup C \cup D}) = 1000 - 564$
$= 436$

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3.

$n(U) = 2500$

Let A be course in C
B " Pascal
C " Networking

$n(A) = 1700$; $n(B) = 1000$; $n(C) = 550$
$n(A \cap B) = 750$; $n(B \cap C) = 275$; $n(A \cap C) = 400$
$n(A \cap B \cap C) = 200$

(i) $n(A \cup B \cup C) = ?$

$= n(A) + n(B) + n(C) - [n(A \cap B) + n(B \cap C) + n(A \cap C)] + n(A \cap B \cap C)$.

$= 1700 + 1000 + 550 - (750 + 275 + 400) + 200$
$= 2025_{//}$

(ii) $n(\overline{A \cup B \cup C}) = ?$

$= 2500 - 2025$
$= 475_{//}$

Note:

qn can be asked to find nums which are not divisible by 2/3/5 & divisible by 7.

find $n(\overline{A \cup B \cup C})$ and to find : $n(\bar{A} \cup \bar{B} \cup \bar{C} \cup D)$
$\hookrightarrow y$

$= \left\lfloor \frac{y}{7} \right\rfloor$

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* RECURRANCE RELATION

An equation that expresses in terms of one or more no. of preceding terms of the sequence.

eg: $\{a_n\}$, $\{F_n\}$, $\{T_n\}$, $\{P_n\}$, $\{S_n\}$

RE : $S_n = S_{n-1} + 1$; soln : $S_n = n$

Ex.

1.

IC : $S_0 = 1$

$S_1 = 2 = 2 \times S_0$

$S_2 = 4 = 2 \times S_1$

$\therefore$ RE : $S_n = 2 \times S_{n-1}$ $\therefore$ soln : $S_n = 2^n$

2. Fibonacci series

IC : $\begin{cases} S_0 = 1 \\ S_1 = 1 \end{cases}$

$S_2 = 2 = S_0 + S_1$

$S_3 = 3 = S_1 + S_2$

RE : $S_n = S_{n-1} + S_{n-2}$ $\therefore$ soln : $f(n)$

3.

RE : $a_n = a_{n-1} + 3$, $n \geq 1$

IC : $a_0 = 2$

$a_1 = a_0 + 3 = 5$

$a_2 = a_1 + 3 = 8$

$a_3 = a_2 + 3 = 11$

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4.

RE : $a_n = a_{n-1} - a_{n-2}$, $n \geq 2$

IC : $a_0 = 3$; $a_1 = 5$

$a_2 = a_1 - a_0 = 2$

$a_3 = a_2 - a_1 = -3$ //

*** SOLVING 1st & 2nd ORDER L.R.E**

- Solution of $a_n = a_h + a_p$

a_h ↓ homogenous soln

a_p ↳ particular soln

- $a_n = c_1 a_{n-1} + c_2 a_{n-2} \cdots c_k a_{n-k} + f(n)$

↳ recurrance relation of order k, where $c_1, c_2 ..$ are real numbers.

- if $f(n) = 0$: homogenous

$f(n) \neq 0$: non-homogenous

- general soln : $a_n = a_h + \boxed{a_p}$ → is zero for homogenous

*** FINDING a_h**

if roots are :

1. $m_1 \neq m_2$: $a_h = Am_1^n + Bm_2^n$
2. $m_1 = m_2 = m$: $a_h = (An + B)m^n$

*** FINDING a_p**

if. f(n) is :

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1. $(b_0 + b_1 n + b_2 n^2 \dots) s^n$: $a_p = (p_0 + p_1 n + p_2 n^2 \dots p_k n^k) s^n$
where $b_0, b_1, s \in R$ & s is not a root.

when s is a root : $a_p = (p_0 + p_1 n + p_2 n^2 \dots p_k n^k) n \cdot s^n$

Ex.

1.

$a_n = 2a_{n-1} + 3^n$

IC : $a_1 = 5$

GS : $a_h + a_p$

$a_n - 2a_{n-1} = 3^n \rightarrow$ 1st order (1)

$(D-2)y$

a_h:

$m - 2 = 0$

$m = 2$, one distinct root

$a_p = A(2)^n$

a_p:

$f(n) = 3^n$, since 3 is not a root

trial soln : $a_p = A \cdot 3^n$, subst. in (1)

$A \cdot 3^n - 2A(3^{n-1}) = 3^n$

$3A - 2A = 3$

$A = 3$

$\therefore a_p = 3 \cdot 3^n = 3^{n+1}$ $\therefore$ GS : $a_n = A \cdot 2^n + 3^{n+1}$

$a_p = P \cdot n \cdot 5^n$ = trial soln, subst in ①

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$$P n(5^n) - 3P(n-1)(5^{n-1}) - 10P(n-2)(5^{n-2}) = 7 \cdot 5^n$$

$a_1 = A \cdot 2^1 + 3^2$

$5 = 2A + 9$

$2A = -4$

$A = -2$ //

$\therefore \quad a_n = -2^{n+1} + 3^{n+1}$ //

$$25P(n) - 3P(n-1)(5) - 10P(n-2) = 7 \cdot 5^n$$

$$\cancel{25P(n)} - \cancel{15Pn} + 15P - \cancel{10P(n)} + 20P = 7 \cdot 5 : 5$$

$$35P = 7 \cdot (25)$$

$$\boxed{P = 5}$$

2.

$f(n) = 3 \cdot f_{n-1} + 10 \cdot f_{n-2} + 7(5^n)$ IC: $f_0 = 4$, $f_1 = 3$.

$f_n - 3f_{n-1} - 10 f_{n-2} = 7 \cdot 5^n \longrightarrow$ ①

a_h:

$m^2 - 3m - 10 = 0$

$m(m-5) + 2(m-5) = 0$

$m = 5 \;/\; m = -2$, distinct

$a_h : A(5)^n + B(-2)^n$

$a_p = n 5^{n+1}$ //

$a_n = A \cdot 5^n + B(-2)^n + n5^{n+1}$

$a_0 = A - B = 4.$

$a_1 = 5A - 2B + 25 = 3$

$5A - 2B = -22.$

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Ex.

1.

$$a_n + 6a_{n-1} + 9a_{n-2} = (-3)^n \quad n \geq 2$$

$\hookrightarrow$ ①

IC: $a_0 = 2$

$a_1 = 3$

soln : $a_n = a_h + a_p$

a_h:

$m^2 + 6m + 9 = 0$

$m = -3/-3$

$a_h = (An + B)(-3)^n$

a_p:

Let trial soln $a_p = a_n = An^2(-3)^n$

Subst in ①

$An^2(-3)^n + 6A(n-1)^2(-3)^{n-1} + 9A(n-2)^2(-3)^{n-2} = (-3)^n$

$An^2(9) + 6A(n-1)^2(-3) + 9A(n-2)^2 = 9$

$A(9n^2 + (-18)(n-1)^2 + 9(n-2)^2) = 9$

$A(n^2 - 2(n^2+1-2n) + (n^2+4-4n)) = 1$

$A(2) = 1$

$\boxed{A = 1/2}$

$\therefore a_p = \dfrac{n^2(-3)^n}{2}$

$a_n = (An+B)(-3)^n + \dfrac{(-3)^n n^2}{2}$

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$a_0 = B = 2.$

$a_1 = (A+B)(-3) + \frac{(-3)}{2} = 3$

$-6(2+A) - 3 = 6$

$-4 - 2A - 1 = 2$

$2A = -7$

$\boxed{A = -7/2}$

$a_n : \left(\frac{-7n}{2} + 2\right)(-3)^n + \frac{(-3)^n n^2}{2}$ //

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* GENERATING FUNCTION

Generating function represents sequences where each term of a sequence is expressed as a coefficient of variable x in a formal power series.

eg: $a_0 + a_1 x + a_2 x^2 \dots = \sum_{n=0}^{\infty} a_n x^n = G(x)$

· $G(x)$ for known sequences:

1. $a_n = \frac{1}{2^n}, \quad n \geq 0$

$$G(x) = \sum_{n=0}^{\infty} \left(\frac{x}{2}\right)^n = (1 - x/2)^{-1} = \frac{2}{2-x}$$

2. $a_n = \frac{1}{n!}, \quad n \geq 0$

$$G(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!} = e^x$$

3. $a_n = a^n, \quad n \geq 0$

$$G(x) = \sum_{n=0}^{\infty} (ax)^n = (1 - ax)^n = \frac{1}{1-ax} \qquad |ax| < 1$$

4. $a_n = (n+1), \quad n \geq 0$

$$G(x) = \sum_{n=0}^{\infty} x^n (n+1) = 1 + 2x + 3x^2 \dots = \frac{1}{(1-x)^2} = (1-x)^{-2}$$

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Ex.

1.

use generating function, solve :

$a_n - 2a_{n-1} = 3^{n-1}, \; n \geq 1$ IC: $a_0 = 5$

$\Rightarrow a_n \cdot x^n - 2 \cdot a_{n-1} \cdot x^n = 3^{n-1} \cdot x^n$

$\Rightarrow \sum_{n=1}^{\infty} a_n \cdot x^n - 2\sum_{n=1}^{\infty} x^n \cdot a_{n-1} = \sum_{n=1}^{\infty} 3^{n-1} \cdot x^n$

$\Rightarrow [a_1 x + a_2 x^2 + a_3 x^3 \ldots] - 2[a_0 x + a_1 x^2 + a_2 x^3 \ldots]$

$= [x + 3x^2 + 9x^3 \ldots]$

$\Rightarrow G(x) - a_0 - 2x\,G(x) = x(1-3x)^{-1}$

$G(x)[1-2x] = \frac{x}{1-3x} + a_0$

$G(x) = 5 + \frac{x}{1-3x} = \frac{5-14x}{1-3x} \cdot \frac{1}{(1-2x)}$

$\therefore G(x) = \frac{5-14x}{(1-3x)(1-2x)}$

$5 - 14x = A(1-2x) + B(1-3x)$

$x = 1/2$

$5 - 7 = -2 = B\left(1 - \frac{3}{2}\right)$

$-4 = -B$

$\boxed{B = 4}$

$x = 1/3$

$5 - \frac{14}{3} = \frac{1}{3} = A\left(1 - \frac{2}{3}\right)$

$\frac{1}{3} = \frac{A}{3}$

$\boxed{A = 1}$

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$$G(x) = \frac{1}{1-3x} + \frac{4}{1-2x}$$

$$G(x) = (1-3x)^{-1} + 4(1-2x)^{-1} \,//$$

$$\therefore \sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} (3x)^n + 4(2x)^n \,//$$

$$\therefore a_n = 3^n + 2^{n+2}, \quad n \geq 0 \,//$$

2.

$$a_n = 2a_{n-1} + 4a_{n-2}, \quad n \geq 2 \qquad IC: a_0 = 1, \; a_1 = 3$$

$$a_n \cdot x^n = 2 \cdot a_{n-1} \cdot x^n + 4 \cdot a_{n-2} \cdot x^n$$

$$\sum_{n=2}^{\infty} a_n \cdot x^n = 2 \sum_{n=2}^{\infty} a_{n-1} x^n + 4 \sum_{n=2}^{\infty} a_{n-2} \cdot x^n$$

$$[a_2 x^2 + a_3 x^3 \ldots] = 2[a_1 x^2 + a_2 x^3 \ldots] + 4[a_0 x^2 + a_1 x^3 \ldots]$$

$$-(a_0 + a_1 x) + G(x) = 2x\,G(x) - 2a_0 x + 4x^2 \cdot G(x)$$

$$G(x)[1 - 2x - 4x^2] = -2a_0 x + a_0 + a_1 x$$

$$= -2x + 1 + 3x$$

$$-G(x)(4x^2 + 2x - 1) = x + 1$$

$$G(x) = \frac{-(x+1)}{(4x^2+2x-1)} = \frac{Ax+B}{4x^2+2x-1}$$

$A = -1, \; B = -1$ //

This is an improper fraction//.

3.

$$a_n = 9a_{n-1} + (10)^{n-1}, \quad n \geq 1 \qquad IC: a_0 = 9$$

$$a_n \cdot x^n = 9a_{n-1} \cdot x^n + 10^{n-1} \cdot x^n$$

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$$\sum_{n=1}^{\infty} a_n x^n - 9 \sum_{n=1}^{\infty} a_{n-1} \cdot x^n = \sum_{n=1}^{\infty} 10^{n-1} \cdot x^n$$

$$[a_1 x + a_2 x^2 \dots] - 9[a_0 x + a_1 x^2 \dots] = [x + 10x^2 + 100x^3 \dots]$$

we know, $G(x) = \sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 \dots$

$$G(x) - a_0 - 9x \cdot G(x) = x \cdot (1-10x)^{-1}$$

$$G(x)[1-9x] = \frac{x}{1-10x} + 9$$

$$= \frac{9-89x}{1-10x}$$

$$G(x) = \frac{9-89x}{(1-10x)(1-9x)} \text{ //}$$

$$9 - 89x = A(1-9x) + B(1-10x)$$

$x = 1/9$

$$9 - \frac{89}{9} = \frac{-8}{9} = B\left(1 - \frac{10}{9}\right)$$

$$\frac{-8}{9} = \frac{-B}{9} \Rightarrow \boxed{B=8}$$

$x = 1/10$

$$9 - \frac{89}{10} = \frac{1}{10} = A\left(1 - \frac{9}{10}\right)$$

$$\frac{1}{10} = \frac{A}{10} \Rightarrow \boxed{A=1}$$

$$G(x) = \frac{1}{1-10x} + \frac{8}{1-9x}$$

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$$G(x) = (1-10x)^{-1} + 8\cdot(1-9x)^{-1}$$

$$\sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} (10x)^n + 8\sum_{n=0}^{\infty} (9x)^n$$

$$\therefore a_n = 10^n + 8\cdot 9^n \,//.$$

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Ex.

1.

$a_n - 6a_{n-1} + 9a_{n-2} = 0 \;,\; n \geq 2$ IC : $a_0 = 1 \;;\; a_1 = 9$

$$a_n \cdot x^n - 6a_{n-1}\cdot x^n + 9\cdot a_{n-2}\cdot x^n = 0$$

$$\sum_{n=2}^{\infty} a_n \cdot x^n - \sum_{n=2}^{\infty} 6\cdot a_{n-1}\cdot x^n + 9\sum_{n=2}^{\infty} a_{n-2}\cdot x^n = 0$$

$$[a_2x^2 + a_3x^3 \ldots] - 6[a_1x^2 + a_2x^3 \ldots] + 9[a_0x^2 + a_1x^3 \ldots] = 0$$

$$G(x) - a_0 - a_1x - 6G(x)\cdot x - 6a_0x + 9x^2\cdot G(x) = 0$$

$$G(x)[1-6x+9x^2] - 1 - 9x - 6x = 0$$

$$G(x) = \frac{1+15x}{1-6x+9x^2}$$

$$= \frac{1+15x}{(1-3x)^2}$$

$1+15x = A(1-3x) + B$

$$G(x) = \frac{-5}{1-3x} + \frac{6}{(1-3x)^2}$$

$x = 1/3$ $x = 0$

$$= -5(1-3x)^{-1} + 6(1-3x)^{-2}$$

$\boxed{6 = B}$ $1 = 6 + A$ $\boxed{A = -5}$

$$G(x) = -5\sum_{n=0}^{\infty} (3x)^n + 6\sum_{n=0}^{\infty} x^n(n+1)$$

Not x^n it should be (3x)^n

$$G(x) = \sum_{n=0}^{\infty} a_n \cdot x^n$$

$$\therefore a_n = (-5)3^n + 6\cdot(n+1)\ldots$$